

WeshFahmna AI | وش فهمنا ؟

Call Center Conversation Intelligence

From Conversations to Clear Insights

- **Maharah GDP | AI Technical Assessment**
- **Presented by Yasir Aladwani**
- **November 2026**

Every Conversation Contains Valuable Insights

Customer-service calls hold valuable information that is slow to analyze manually. Our goal: transform raw audio into structured, understandable insights.



01 Transcription

Convert speech into text.

02 Speakers

Speaker identification: distinguish customer and agent turns.

03 Summarization

AI summarization: extract issues, actions, and outcomes.

04 Sentiment

Sentiment analysis: understand customer language and emotional trends.

Research-Driven Dataset Selection

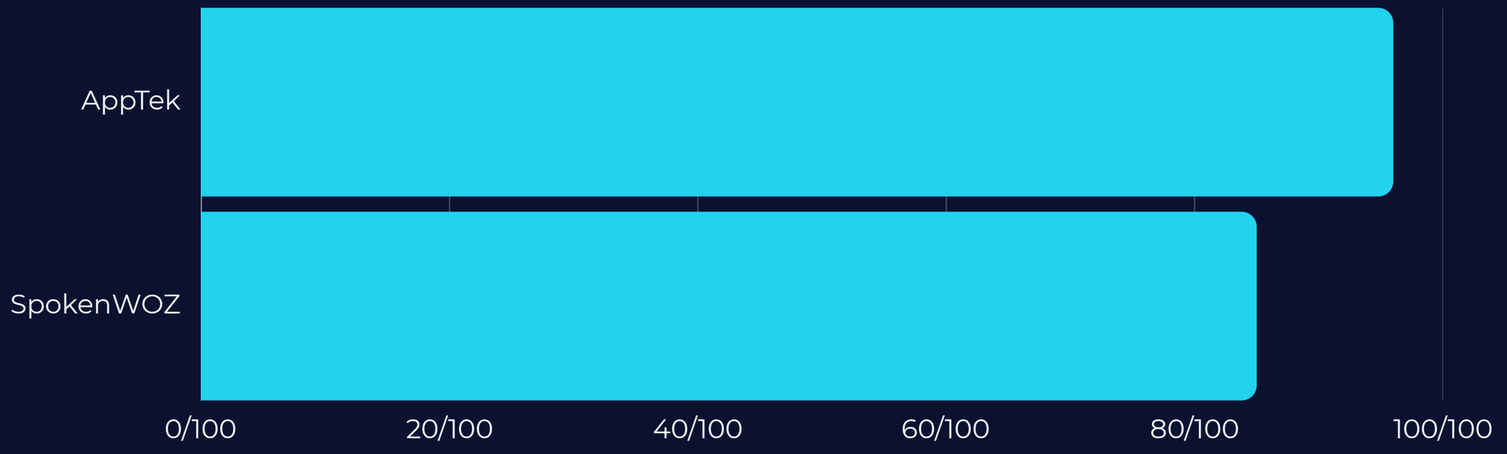
Structured comparison of datasets and research literature before implementation.

16
Datasets Reviewed

15
Papers & Articles

96/100
AppTek Suitability Score

Selected dataset: AppTek Call-Center Dialogues



Why AppTek

- Agent-customer conversations
- Reference transcripts and speaker labels
- Multiple accents and service domains
- Timestamped annotations for evaluation

Project-specific dataset suitability scores, not official benchmarks or model accuracy.

Understanding the Data

873
Conversations — full dataset

12 Calls
Our evaluation pilot

118.83
Minutes of Audio

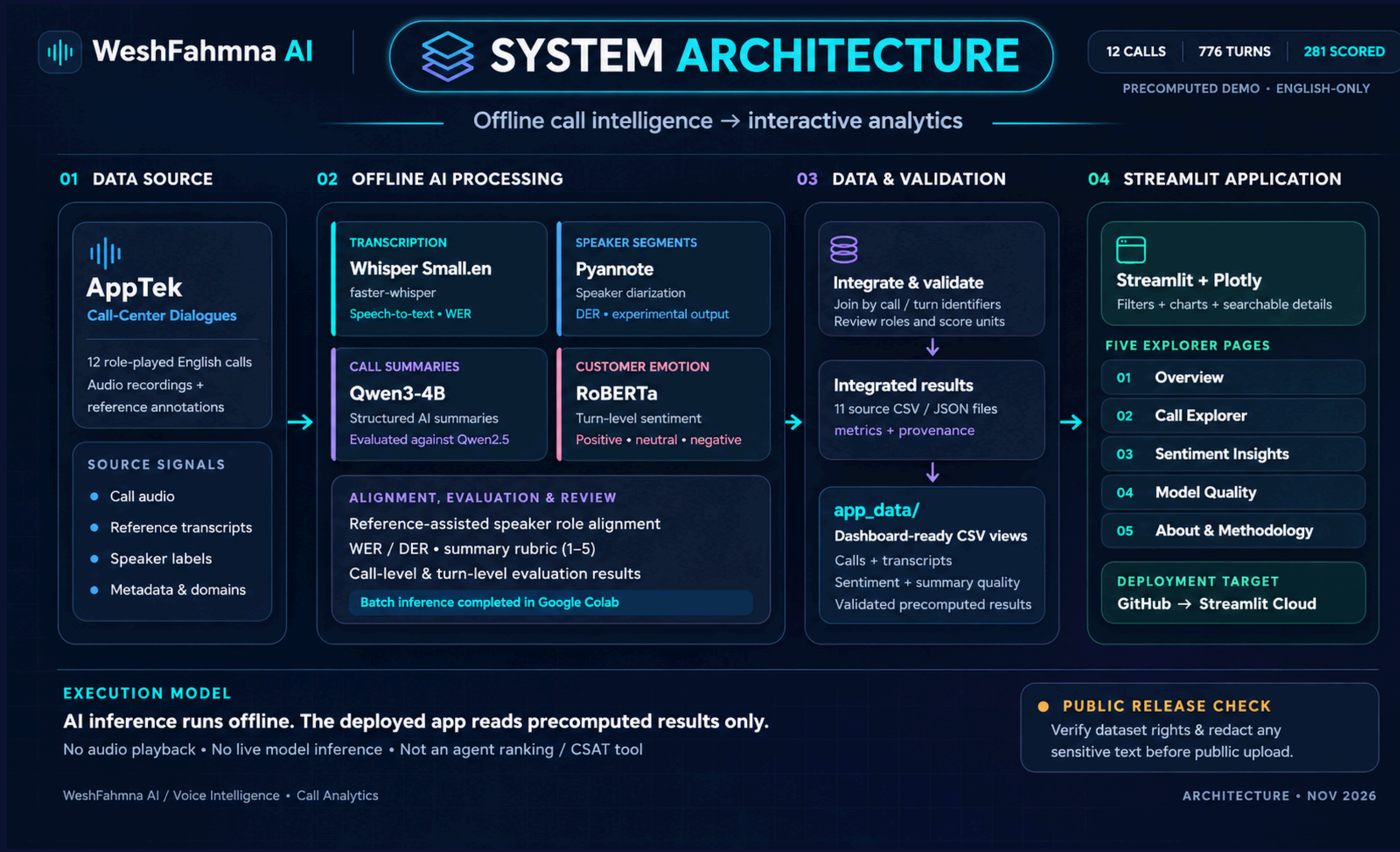
8
Service Domains

4
Accent Groups

776
Speaker Turns

English-language, role-played customer-service conversations with reference annotations.

How WeshFahmna AI Works



Evaluating Speech Understanding

20.85%

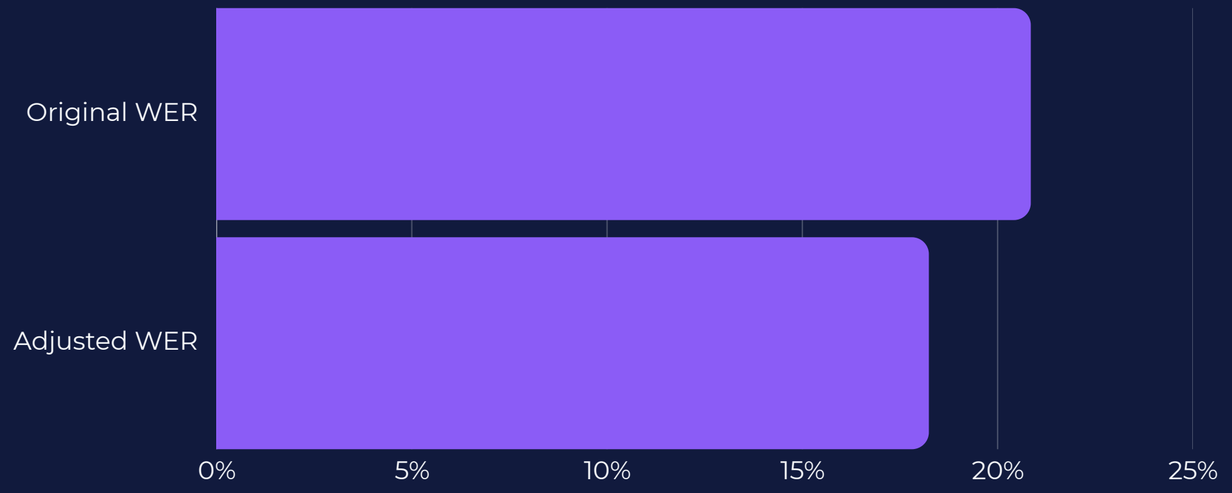
Mean Word Error Rate

18.24%

Filler-Adjusted WER

19.93%

Mean Diarization Error Rate — speaker diarization



2.61 percentage points

Filler normalization reduced reported WER; speaker attribution remained challenging in some conversations.

Average call-level errors across the 12-call pilot. Lower is better. Filler normalization changes the evaluation calculation, not the ASR model.

Turning Conversations into Structured Summaries

LONG CONVERSATION
↓
Customer Issue
Agent Action
Final Outcome / Next Step

3.31 / 5

Qwen2.5-3B-Instruct average quality

3.83 / 5

Qwen3-4B-Instruct-2507 average quality

Head-to-head

7 calls

Qwen3 better

4 calls

Qwen2.5 better

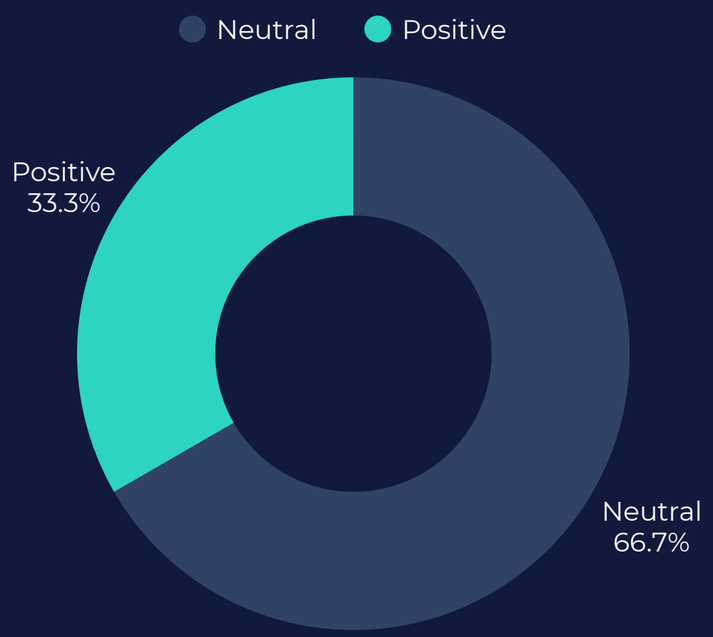
1 tie

Equal review score

Qwen3 achieved stronger average summary quality in our pilot. Project-specific review-rubric scores, not standardized benchmarks.

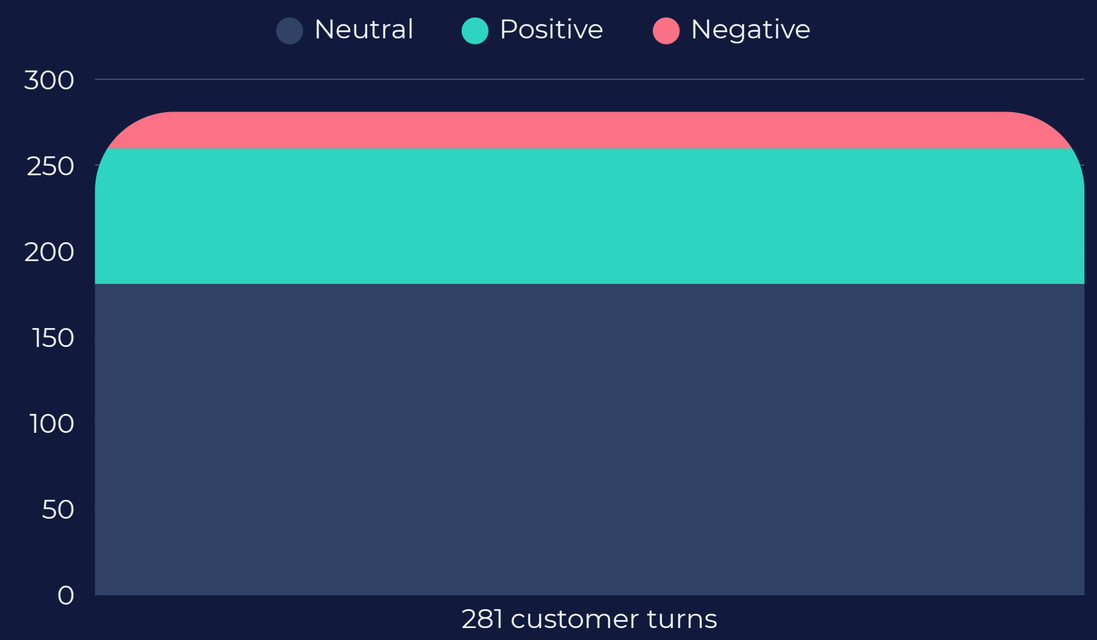
Understanding Customer Sentiment

Overall sentiment across 12 calls



8 Neutral • 4 Positive • 0 Negative

Customer-turn sentiment



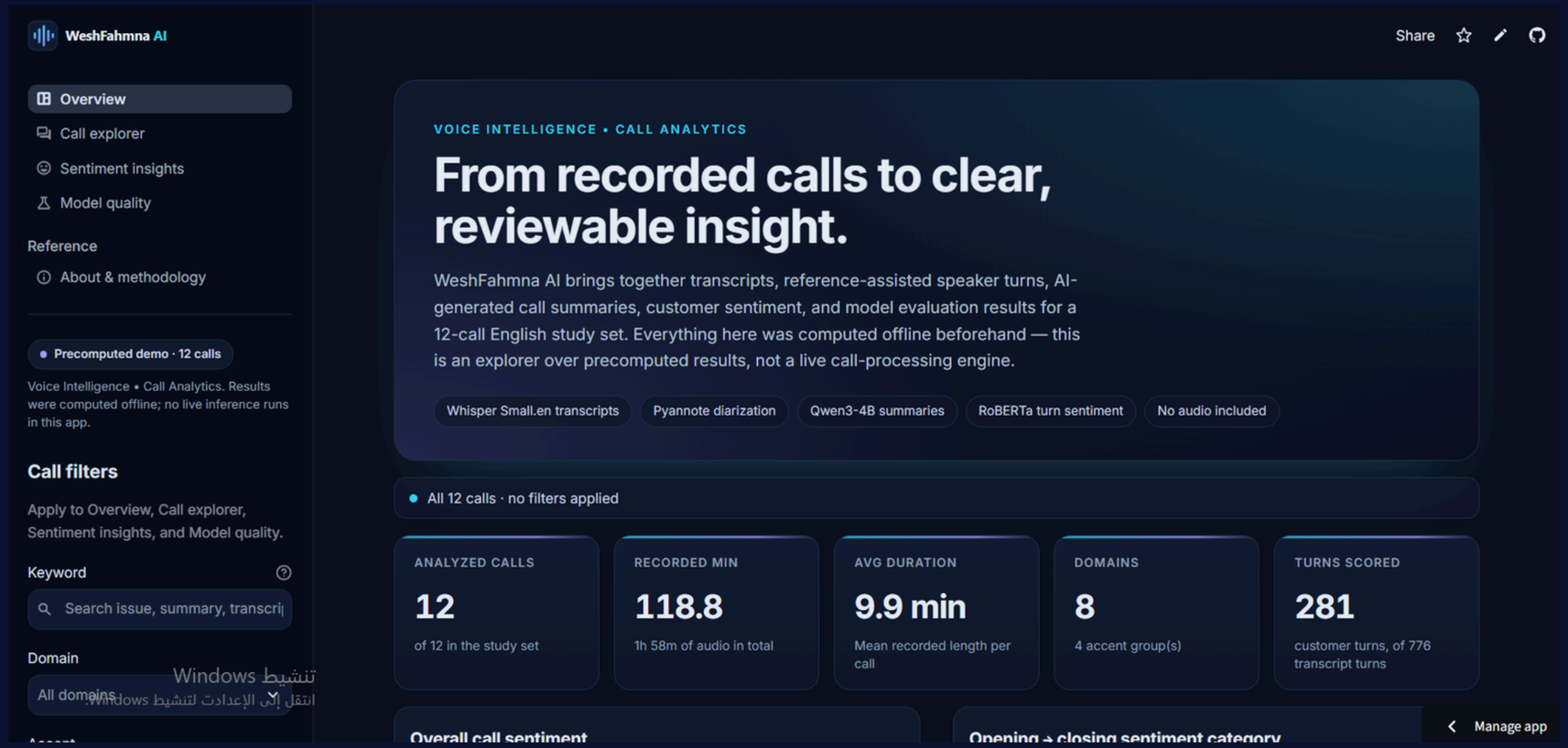
181 Neutral • 79 Positive • 21 Negative

Negative customer moments appeared even in calls classified neutral or positive overall.

Sentiment reflects model-classified language, not verified customer satisfaction.

From AI Models to a Working Application

WeshFahmna AI Dashboard — an interactive Streamlit application for exploring conversation insights and model evaluations.



Executive Overview | **Call Explorer** | **Sentiment Insights** | **Model Quality** | **Methodology**

LIVE DEMO • weshfahmna-ai.streamlit.app

What We Learned

01 Speech Recognition Errors

Accents, fillers, and natural speech affected transcription quality.

Lesson: Measure accuracy against reference transcripts.

02 Speaker Attribution

Some conversations produced significant diarization errors.

Lesson: Keep predicted labels separate from reference-assisted roles.

03 LLM Reliability

Generated summaries occasionally misinterpreted details or outcomes.

Lesson: Evaluate factual accuracy and preserve uncertainty.

Reliable AI requires validation, transparency, and human oversight.





Thank You

An end-to-end AI prototype combining Speech Recognition, Speaker Analysis, LLM Summarization, Sentiment Analytics and Interactive Visualization.

Future improvements: expand evaluation beyond 12 calls; improve independent speaker attribution; strengthen factual verification of AI summaries.

Live Application • weshfahmna-ai.streamlit.app

GitHub • github.com/Yasir-Aladwani/WeshFahmna-AI

Yasir Aladwani • Maharah GDP • November 2026